What sets the MutantXL closure degree: not N; ℓ = 7 or m·ℓ ≥ 21 remain

These are the results of `research/dreg_closure_driver_20261016/`. The full text is in `RESULTS.md`. **Class:** stage diagnostic. The `n = 131` statements are **derived**.

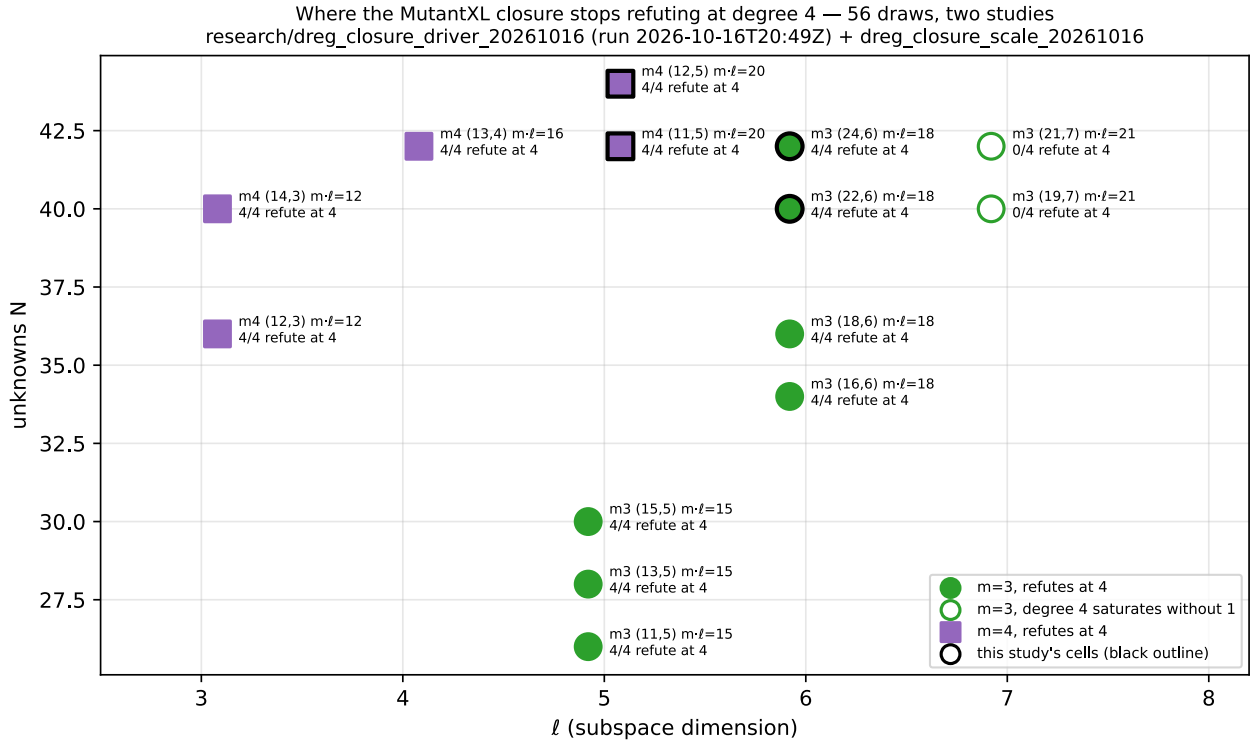


Figure 1: 56 unsatisfiable draws over two studies, placed by (ℓ, N). This study's four cells (16 draws) are outlined in black.

Prediction	outcome	confidence
Q7: m = 3, ℓ = 6 refutes at 4 at N = 40, 42	**holds**: 8 of 8	about 70%
Q8: m = 4, m·ℓ = 20 rises past 4	**refuted**: 8 of 8 refute at 4	about 50%

- **N:** rejected.
- **m·ℓ with a threshold ≤ 20:** rejected.
- **Surviving explanations:** the degree rises at ℓ = 7, or once m·ℓ ≥ 21. Both fit all 56 draws.
- **ECC2K-130 (derived):** both surviving explanations put the m = 4 rise at ℓ = 6 or 7, far below the priced ℓ = 33. The m = 4 "4, constant" row (2^68.43) is therefore contradicted, and the degree-5 rows (m = 4 at 2^71.13) become the optimistic case.
- **Correction 2026-10-30.** The m = 4 degree-5 row (2^71.13, +10.32) is priced at w = 1, a floor (lower bound), not an optimistic estimate. At the fitted w (degree 5: 2.83, 95% CI 2.59 to 2.92) it is **+61.45 bits over rho in rho-step units** (bootstrap 55.8 to 64.0; degree-4 proxy +52.21), and a structured / block-Wiedemann elimination prices it at about +43. The interval excludes the extrapolation from about 2^16 to 2^36 columns. Source: `research/m4_matrix_profile_20261030/README.md`; `RESEARCH_ECC2K130_DESCENT_DEGREE.md` §7 correction.
- **Next:** m = 4, ℓ = 6 separates the two survivors, and degree 5 past ℓ = 7 decides the rows. Both need more memory or a sparse closure.